

12(a)(i)	cannot predict when a particular nucleus will decay or cannot predict which nucleus will decay next	B1
12(a)(ii)	(decay is) not affected by external (environmental) factors	B1
12(b)(i)	$A = A_0 \exp(-\lambda t)$ and so $\ln A = \ln A_0 - \lambda t$ gradient of line = $(-)\lambda$	C1
	$\lambda = (36.4 - 35.0) / (20 - 0)$ (= 0.07(0) min ⁻¹)	C1
	half-life = $\ln 2 / \lambda$ = $\ln 2 / 0.070$ = 10 min	A1
	or	
	$A_0 = \exp(-36.4) = 6.43 \times 10^{15}$ (Bq)	(C1)
	$A_0 / 2 = 3.21 \times 10^{15}$ (Bq), so $\ln(A_0 / 2) = 35.7$	(C1)
	read off half-life = 10 min	(A1)
	or	
	(at one half-life,) $\ln A = 36.4 - \ln 2$	(C1)
	= 35.7	(C1)
	read off half-life = 10 min	(A1)

Question	Answer	Marks
12(b)(ii)	$A = \lambda N$	C1
	$N = \text{mass} / (\text{nucleon number} \times u)$ or $N = (\text{mass} / \text{nucleon number}) \times N_A$	C1
	$\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-7}) / (\text{nucleon number} \times 1.66 \times 10^{-27})$ or $\exp(36.4) = (1.17 \times 10^{-3} \times 5.66 \times 10^{-4} \times 6.02 \times 10^{23}) / \text{nucleon number}$ nucleon number = 62	A1